

COURSE TITLE : ORGANIC TECHNOLOGY
COURSE CODE : 5072
COURSE CATEGORY : A
PERIODS/ WEEK : 4

PERIODS/ SEMESTER : 52
CREDIT : 4

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Oils and fats, soaps & detergents	13
2	Pulp and Paper	13
3	Explosives and insecticides	13
4	Sugar, starch and leather	13
TOTAL		52

COURSE OUTCOMES :

SL.NO	SUB	STUDENT WILL BE ABLE TO
1	1	Understand the chemistry of fats and oils
	2	Understand the refining of vegetable oils
	3	Understand the manufacture of soap & detergents
2	4	Understand the manufacture of Pulp & Paper.
3	5	Understand the properties and manufacturing of explosives and insecticides
4	6	Understand the properties and process of manufacturing of sugar, starch & leather

SPECIFIC OUTCOMES

MODULE – I OILS AND FATS, SOAPS AND DETERGENTS

1.1.0 Understand the chemistry of fats and oils

- 1.1.1 Distinguish between fats and oils
- 1.1.2 Define saturated and unsaturated fatty acids
- 1.1.3 Define iodine value, saponification value and acid value of oils
- 1.1.4 Distinguish between drying and non-drying oils
- 1.1.5 Define rancidity of oil
- 1.1.6 Select extraction processes for oils and fats
- 1.1.7 List six important raw materials available in India for oil manufacture
- 1.1.8 Describe the extraction process of oil by expellers
- 1.1.9 Describe solvent extraction
- 1.1.10 Differentiate the merits and demerits of extraction by expeller and solvent extraction

1.2.0 Understand the refining of vegetable oils

- 1.2.1 Describe neutralization of oils
- 1.2.2 Describe bleaching of oils
- 1.2.3 Describe deodourisation of oils
- 1.2.4 Draw the flow diagram for the refining of oil
- 1.2.5 State the purpose of hydrogenation process
- 1.2.6 Describe the hydrogenation
- 1.2.7 Identify the catalysts used for hydrogenation

1.3.0 Understand the manufacturing of soap

- 1.3.1 Define soap
- 1.3.2 Differentiate toilet soap and laundry soap
- 1.3.3 Explain with a flow diagram for the manufacture of soap by full boiled process.
- 1.3.4 State different raw materials for soap manufacture
- 1.3.5 Differentiate batch and continuous process
- 1.3.6 Describe batch process for soap manufacture
- 1.3.7 Describe continuous process for toilet soap manufacture
- 1.3.8 State different types of additives added to soap
- 1.3.9 List the importance of perfumes in soap

1.4.0 Understand the importance of spent lye recovery

- 1.4.1 Describe the treatment of spent lye by
- 1.4.2 Neutralization
- 1.4.3 Evaporation
- 1.4.4 Concentration
- 1.4.5 Describe refining of crude glycerine

1.5.0 Understand the importance of synthetic detergents

- 1.5.1 List the uses of detergents
- 1.5.2 Define detergency
- 1.5.3 Describe the mechanism of detergents
- 1.5.4 Differentiate anionic, catatonic and non-ionic Detergents.
- 1.5.5 Describe the manufacture of Dodecyl benzene
- 1.5.6 Describe the manufacture of Alkyl benzene Sulphonate (ABS)

MODULE – II PULP AND PAPER

2.1.0 Understand the manufacture of Pulp

- 2.1.1 List the raw materials for pulp making
- 2.1.2 Describe the preparations of raw materials for pulping
- 2.1.3 Describe manufacture of pulp from waste paper
- 2.1.4 Distinguish chemical, mechanical and semi chemical process for pulp manufacture
- 2.1.5 Draw the flow diagram for sulphate process
- 2.1.6 Describe sulphate process
- 2.1.7 Describe soda process for pulping
- 2.1.8 Describe the different methods of bleaching using different bleaching agent

2.2.0 Understand the manufacturing of paper

- 2.2.1 Describe preparation of pulp for paper manufacture
- 2.2.2 List the additives mixed with pulp
- 2.2.3 Explain the purpose of using additives
- 2.2.4 Draw a line diagram of Fourdiner paper making machine
- 2.2.5 Describe the working of paper making machines
- 2.2.6 Describe the manufacture of laminated paper
- 2.2.7 List the advantages of laminate paper

MODULE-III EXPLOSIVES AND INSECTICIDES

3.1.0 Understand the characteristics of explosives

- 3.1.1 Define and explain sensitivity, power and permissibility
- 3.1.2 Differentiate properties of primary and secondary explosives
- 3.1.3 Differentiate military and commercial explosives
- 3.1.4 Describe high explosive ammunition
- 3.1.5 Draw a flow diagram for the manufacture of nitrocellulose
- 3.1.6 Describe the manufacture of nitrocellulose and smokeless powder
- 3.1.7 Describe briefly the manufacture of PETN, RDX & Lead Oxide
- 3.1.8 Describe the manufacture of nitro-glycerine and Dynamite

3.2.0 Understand the properties and manufacturing of insecticides

- 3.2.1 State the important properties of insecticides
- 3.2.2 Describe the mode of action of insecticides with examples
- 3.2.3 Describe the manufacture of Bio Insecticides.
- 3.2.4 List the advantages of Bio Insecticides over other insecticides.

MODULE – IV SUGAR, STARCH, AND LEATHER

4.1.0 Understand the properties and manufacturing of cane sugar

- 4.1.1 Describe the different steps involved in the preparation of cane for milling
- 4.1.2 Describe milling and extraction of cane juice
- 4.1.3 Describe purification of cane juice for evaporation
- 4.1.4 Describe evaporation and crystallization of juice for the manufacture of sugar
- 4.1.5 Explain centrifuging and drying processes in sugar manufacture.
- 4.1.6 Describe refining process of sugar
- 4.1.7 Describe utilization of by-products from sugar mill

4.2.0 Understand manufacturing of starch

- 4.2.1 Describe the manufacture of starch from Tapioca and Corn
- 4.2.2 Describe bleaching, centrifuging and drying of starch

4.3.0 Understand conversion of skin into leather

- 4.3.1 Describe preparation of skin and hide for tanning
- 4.3.2 Describe vegetable tanning and chrome tanning processes
- 4.3.3 Describe finishing operations of leather

COURSE CONTENT

MODULE – I

Oils and Fats

Chemistry of saturated and unsaturated fats and oils – drying and non drying oils – causes of rancidity – acid value, saponification value and iodine value. Extraction of oils – oil expellers and solvent extraction. Advantages and disadvantages of each. Refining of vegetable oils, neutralization, bleaching and deodourisation. Hydrogenation – Purpose of hydrogenation, chemistry and process details, preparation of catalysts

Soaps and detergents – study of different types of toilet soaps and washing soaps. Raw materials, manufacturing process for toilet & laundry soaps. Batch and continuous process, glycerine recovery, different types of fillers and perfumes

Chemistry of important detergents, Alkyl aryl sulphonate, alkyl sulphonates – condensates, comparison with soap. Biodegradability of detergents, detergent additives and formulation for liquid and powdered detergent

MODULE – II

Pulp & Paper

Paper – Paper industry in India and its future.

Manufacture of pulp and paper – Raw materials – Chemistry of wood pulping process. Sulphate, Soda mechanical pulping – details of process and plant.

Treatment of pulp: – Bleaching – sizing, loading, filling, colouring, beating, and refining of pulp for paper manufacture. Details of paper machine

MODULE-III

Explosives and Insecticides

Explosives – General characteristics of Industrial and military explosives – dynamite, nitro- glycerine, cellulose nitrate, smokeless powder, trinitrotoluene manufacture.

Insecticides – Classifications – Inorganic – organic and natural – classification based on mode of action – inorganic – sulphur compounds – phosphorus compounds – chemistry – mode of action and application – manufacture and process details of Bio insecticides.

MODULE – IV

Sugar, Starch and Leather

Sugar: - Manufacture of raw cane sugar – composition of cane and cane juice, chipping, shredding and classification by liming, evaporation, crystallization and centrifuging. Refined sugar, bleaching, concentration, crystallization, centrifuging and drying. Direct consumption sugar – Extraction, carbonation and sulphitation, filtration, concentration, vacuum operation centrifuging and drying, modern development in refining sugar, utilization of by-products (molasses and Baggase).

Starch: - Manufacture – from tapioca and corn, steeping disintegrating –bleaching, filtration and drying. Different grades of starch – bleaching.

Leather: - Animal skin – preparation of raw hides and skins, vegetable tanning, chrome tanning, synthetic tanning agents, and finishing operations

REFERENCES

- James.E.Bailey and David.F.Olis - Biochemical Engineering Fundamentals
- Dryden - Out lines of chemical Technology
- Shreve - Chemical Process Industries